

Bachelor of Science (Honours) in Data Science and Artificial Intelligence

DA 102 – Reference Functions

Learning Objectives



- 01** Referring cells using VLOOKUP, HLOOKUP
- 02** Referring cells using XLOOKUP
- 03** Referring cells using INDEX
- 04** Referring cells using MATCH
- 05** Analytics questions involving these functions





Referring Cells using VLOOKUP



Referring Cells Using VLOOKUP - 01



- Individual Cells are referred using (column name, row number) format
- One must know the column name AND row number to access a cell.
- Many a times we know a value and we want to access cells based on the known value.

Referring Cells Using VLOOKUP - 02



- Roll numbers are stored in column A
- Names are stored in column B
- CPI information is stored in column C
- Given a roll number (say 124), access corresponding CPI of the student.
- CPI is not accessed using C3 however with the known value of roll number.
- **Cell value** to be accessed is present in C3.

	A	B	C
1	Roll Number	Name	CPI
2	123	Alok Kumar	7.2
3	124	Anmol	8.4
4	125	Amar Nath	9.6

Referring Cells Using VLOOKUP - 03



- The cell information to be accessed need not be a complete table such as A1: C10
- A search between row 5 and row 10 (red color) alone to can be performed. That is A5: C10

	A	B	C
1	Roll Number	Name	CPI
2	123	Alok Kumar	7.2
3	124	Anmol	8.4
4	125	Amar Nath	9.6
5	126	Arogya	7.1
6	127	Arghya	7.2
8	128	Venkat	8.3
9	129	Keshav	9.5
10	130	Sachin	10.0

Referring Cells Using VLOOKUP - 04



- Employee numbers are stored in column A
- Role of the employee is stored in column B
- Project on which the employee is working is stored in column C.
- Given the employee number (say 124), access the project number the employee is working (P101).
- In this example, cell is not accessed using (column name, row number) method such as C3.
- The known information is employee number. This information can be accessed using A3.

	A	B	C
1	Employee Number	Role	Project Number
2	123	Senior Manager	P101
3	124	Developer	P101
4	125	Senior Developer	P102

Referring Cells Using VLOOKUP - 05



- In each of these examples, given a value, retrieve information that we are seeking.
- VLOOKUP help achieve this task
- VLOOKUP is a function.
- It takes FOUR arguments
- Returns the value that is being associated with the search value.
 - Value that is being searched
 - Example: 124 (roll number)
 - The range of cells in which the value is being searched
 - A2:C4 (3 cells in column A, Column B and Column C). One of the columns must correspond to roll numbers
 - Column index from which the search value is being obtained. In this example, column index is 3
 - Range look up: TRUE or FALSE.

Referring Cells Using VLOOKUP - 06



- **Lookup Value** that is being searched – example roll number 124
- **The range of cells** in which the value is being searched
 - A2:C4 (4 cells in columns A, B and C). One of the columns must correspond to roll numbers
- **Column index** from which the search value is being obtained
- **Optional: TRUE OR FALSE**
 - For an exact match – FALSE
 - For an approximate match - TRUE.

Referring Cells Using VLOOKUP - 07



Exact Match

- Given the roll number (say 124), access the CPI of the student.
- **VLOOKUP(A3, A2:C4, 3, FALSE)**
- **FALSE** stand for exact match
- Excel file: **lecture04-examples.xlsx**;

	A	B	C
1	Roll Number	Name	CPI
2	123	Alok Kumar	7.2
3	124	Anmol	8.4
4	125	Amar Nath	9.6

Referring Cells Using VLOOKUP – 08



Exact Match

	A	B
	Roll Number	Grade
1	123	AB
2	124	AA
3	125	BB
4	126	BC
5	127	CC

Given the roll number 125, retrieve the grade.

Roll number is in column A, Grade is in column B

VLOOKUP input

- 125 (roll number whose grade we are searching)
- Group of cells: A1:B5
- Column number from where the value to be retrieved
- Exact match or approximate match
- `VLOOKUP(A3, A1:B5, 2, FALSE)`

Referring Cells Using VLOOKUP - 09



Exact Match

- Given the roll number say 125
- Get the **Quiz 1** marks of roll number 125
- **VLOOKUP(A4, A2:F5, 2, FALSE)**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93

Referring Cells Using VLOOKUP - 10



Exact Match

- Given the roll number say 125
- Get the **Quiz 2** marks of roll number 125
- **VLOOKUP(A4, A2:F5, 3, FALSE)**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93



Referring Cells Using VLOOKUP - 11

Exact Match

- Given the roll number say 125
- Get the **Mid Sem** marks of roll number 125
- **VLOOKUP(A4, A2:F5, 4, FALSE)**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93



Referring Cells Using VLOOKUP - 12

Exact Match

- Given the roll number say 125
- Get the **End Sem** marks of roll number 125
- **VLOOKUP(A4, A2:F5, 5, FALSE)**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93



Referring Cells Using VLOOKUP - 13

Exact Match

- Given the roll number say 125
- Get the **Total** marks of roll number 125
- **VLOOKUP(A4, A2:F5, 6, FALSE)**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93

Referring Cells Using VLOOKUP - 14



Exact Match

- Given the roll number that does not exist
- Get the **Total** marks of roll number
- **VLOOKUP(A6,A2:F5, 6, FALSE)**
- **Will result in #N/A error**

	A	B	C	D	E	F
1	Roll Number	Quiz 1	Quiz 2	Mid sem	End Sem	Total
2	123	10	12	24	39	85
3	124	9	11	23	38	81
4	125	11	13	25	40	89
5	126	12	14	26	41	93

Referring Cells Using VLOOKUP - 15



Approximate Match

- We look at examples involving approximate match.
- When the value being searched is not present in the first column of the table, then Excel bases the lookup on **the largest value in the first column that is less than the lookup value**

Referring Cells Using VLOOKUP - 16



Approximate Match

	C	D	E	F	G
3	Lookup Tables				
4					
5		Income	Tax rate		
6		0	0.15		
7		10000	0.3		
8		30000	0.34		
9		100000	0.4		
10					
11					
12		Income	Rate		
13		-1000	Err:504	=VLOOKUP(D13, D6:E9, 2, 1)	
14		30000	0.34	=VLOOKUP(D14, D6:E9, 2, 1)	
15		29000	0.3	=VLOOKUP(D15, D6:E9, 2, 1)	
16		98000	0.34	=VLOOKUP(D16, D6:E9, 2, 1)	
17		104000	0.4	=VLOOKUP(D17, D6:E9, 2, 1)	
18					
19					

When Income is –1000:

Excel searches for largest value in D6:D9 that is less than –1000.

As all values in D6 to D9 are greater than –1000, the formula return an error

Referring Cells Using VLOOKUP - 17



Approximate Match

When Income is 30000:

As there is an exact match found, the formula return 0.34

	C	D	E	F	G
3	Lookup Tables				
4					
5		Income	Tax rate		
6		0	0.15		
7		10000	0.3		
8		30000	0.34		
9		100000	0.4		
10					
11					
12		Income	Rate		
13		-1000	Err:504	=VLOOKUP(D13, D6:E9, 2, 1)	
14		30000	0.34	=VLOOKUP(D14, D6:E9, 2, 1)	
15		29000	0.3	=VLOOKUP(D15, D6:E9, 2, 1)	
16		98000	0.34	=VLOOKUP(D16, D6:E9, 2, 1)	
17		104000	0.4	=VLOOKUP(D17, D6:E9, 2, 1)	
18					
19					

Referring Cells Using VLOOKUP - 18



Approximate Match

When Income is 29000:

Excel searches for largest value in D6:D9 that is less than 29000 which is 10000.

The tax rate corresponding to 10000 is 0.30 which is retrieved.

	C	D	E	F	G
3	Lookup Tables				
4					
5		Income	Tax rate		
6		0	0.15		
7		10000	0.3		
8		30000	0.34		
9		100000	0.4		
10					
11					
12		Income	Rate		
13		-1000	Err:504	=VLOOKUP(D13, D6:E9, 2, 1)	
14		30000	0.34	=VLOOKUP(D14, D6:E9, 2, 1)	
15		29000	0.3	=VLOOKUP(D15, D6:E9, 2, 1)	
16		98000	0.34	=VLOOKUP(D16, D6:E9, 2, 1)	
17		104000	0.4	=VLOOKUP(D17, D6:E9, 2, 1)	
18					
19					

Referring Cells Using VLOOKUP - 19



Approximate Match

When Income is 98000:

Excel searches for largest value in D6:D9 that is less than 98000 which is 30000.

The tax rate corresponding to 30000 is 0.34 which is retrieved.

	C	D	E	F	G
3	Lookup Tables				
4					
5		Income	Tax rate		
6		0	0.15		
7		10000	0.3		
8		30000	0.34		
9		100000	0.4		
10					
11					
12		Income	Rate		
13		-1000	Err:504	=VLOOKUP(D13, D6:E9, 2, 1)	
14		30000	0.34	=VLOOKUP(D14, D6:E9, 2, 1)	
15		29000	0.3	=VLOOKUP(D15, D6:E9, 2, 1)	
16		98000	0.34	=VLOOKUP(D16, D6:E9, 2, 1)	
17		104000	0.4	=VLOOKUP(D17, D6:E9, 2, 1)	
18					
19					

Referring Cells Using VLOOKUP - 20



Approximate Match

When Income is 104000:

Excel searches for largest value in D6:D9 that is less than 104000 which is 100000.

The tax rate corresponding to 100000 is 0.4 which is retrieved.

	C	D	E	F	G
3	Lookup Tables				
4					
5		Income	Tax rate		
6		0	0.15		
7		10000	0.3		
8		30000	0.34		
9		100000	0.4		
10					
11					
12		Income	Rate		
13		-1000	Err:504	=VLOOKUP(D13, D6:E9, 2, 1)	
14		30000	0.34	=VLOOKUP(D14, D6:E9, 2, 1)	
15		29000	0.3	=VLOOKUP(D15, D6:E9, 2, 1)	
16		98000	0.34	=VLOOKUP(D16, D6:E9, 2, 1)	
17		104000	0.4	=VLOOKUP(D17, D6:E9, 2, 1)	
18					
19					



Referring Cells Using VLOOKUP - 21

- **VLOOKUP** searches for values in the righthand direction
- **Lefthanded search** cannot be performed using **VLOOKUP** alone



	A	B	C	D	E	F
1	Quiz 1	Quiz 2	Mid sem	End Sem	Total	Roll Number
2	10	12	24	39	85	123
3	9	11	23	38	81	124
4	11	13	25	40	89	125
5	12	14	26	41	93	126

Referring Cells Using VLOOKUP - 22



- When duplicate values are present, VLOOKUP will find the first match.

	A	B
1	Blue	15
2	Blue	19
3	Green	17
4	Green	14
5	Green	16
6	Red	11
7	Red	18



=VLOOKUP(A3,A1:B7,2,FALSE)

Referring Cells Using VLOOKUP - 23



- First column unsorted- perform exact search

	H	I	J	K
1				
2				
3				
4				
5				
6				
7				
8				
9			Lookup2=H11:I15	
10	Product ID	Price		
11	A134	\$ 3.50		
12	B242	\$ 4.20		
13	X212	\$ 4.80		
14	C413	\$ 5.00		
15	B2211	\$ 5.20		
16				
17	ID	Price		
18	B2211			
19	B2211			

=VLOOKUP(H18,H11:I15,2,FALSE)





Referring Cells using HLOOKUP



Referring Cells Using HLOOKUP - 01



- Instead of looking up values in columns, HLOOKUP look up for values in rows.
- HLOOKUP is a function
- It takes FOUR arguments
- Returns the value that is being associated with the search value.
 - Value that is being searched
 - Example: 04/01/2005
 - The range of cells in which the value is being searched
 - B2:D3 (2 cells in column B, Column C and Column D). One of the columns must correspond to the above date
 - **Row index** from which the search value is being obtained. In this example, it is 2nd row.
 - Range look up: TRUE or FALSE.

Referring Cells Using HLOOKUP - 02



- Let the price of a product depends on date when the product was sold.
- Task is to write a formula to find product price for any date on which product was sold

B12						
	A	B	C	D	E	F
1	Lookup Table					
2	Date	1/1/2005	5/1/2005	8/1/2005		
3	Price	98	105	112		
4						
5			Lookup:B2:D3			
6		Input	Result			
7		Date	Price			
8		1/4/2005	98	=HLOOKUP(B8,B2:D3, 2, TRUE)		
9		5/10/2005	105	=HLOOKUP(B9,B2:D3, 2, TRUE)		
10		9/10/2005	112	=HLOOKUP(B10, B2:D3, 2, TRUE)		
11		5/1/2005	105	=HLOOKUP(B11, B2:D3, 2, TRUE)		

HLOOKUP(B8, B2:D3, 2, TRUE)





XLOOKUP Function



Referring Cells Using XLOOKUP - 01



- **XLOOKUP work with column data (as in VLOOKUP), row data (as in HLOOKUP)**
- **Can lookup both righthand direction and lefthand direction**
- **Can return more than one value**
- **Can perform reverse search**

Referring Cells Using XLOOKUP - 02



- **XLOOKUP arguments**
- **Lookup value** – a cell OR a group of cells
- **Lookup array** – the range of cells within which search should be performed
- **Return array** – Column/row from where values to be returned
- **Not found** – Value to return if no match is found
- **Match mode** - {Exact, Exact or Next smallest, Exact or Next Largest, ...}
- **Search mode** – search from {first, last, ...}

Referring Cells Using XLOOKUP - 03



A1		Horizontal lookup - one value lookup at a time									
	H	J	K	L	M	N	O	P	Q		
1	Vertical Lookup - one value lookup at a time	Lookup Table									
2					Mid se	End			Look =XLC		
3		Roll Number	Quiz 1	Quiz 2	m	Sem	Total				
4		123	10	12	24	39	85				
5		124	9	11	23	38	81				
6		125	11	13	25	40	89				
7		126	12	14	26	41	93				
8		Input the lookup values one by one									
9		Input	Result								
10		J3	85	=XLOOKUP(J3, J3:J6,O3:O6,"NA",0,1)							
11		J4	81	=XLOOKUP(J4, J3:J6,O3:O6,"NA",0,1)							
12		J5	89	=XLOOKUP(J5, J3:J6,O3:O6,"NA",0,1)							
	J6	93	=XLOOKUP(J6, J3:J6,O3:O6,"NA",0,1)								

Lookup for single value say J3
 =XLOOKUP(J3, J3:J6, O3:O6, "NA", 0, 1)

Referring Cells Using XLOOKUP - 04



A1							
	H	J	K	L	M	N	O
1	Vertical Lookup - one value lookup at a time	Lookup Table					
2		Roll Number	Quiz 1	Quiz 2	Mid se m	End Sem	Total
3		123	10	12	24	39	85
4		124	9	11	23	38	81
5		125	11	13	25	40	89
6		126	12	14	26	41	93
7		Input the lookup values one by one					
8		Input	Result				
9		J3	85	=XLOOKUP(J3, J3:J6,O3:O6,"NA",0,1)			
10		J4	81	=XLOOKUP(J4, J3:J6,O3:O6,"NA",0,1)			
11	J5	89	=XLOOKUP(J5, J3:J6,O3:O6,"NA",0,1)				
12	J6	93	=XLOOKUP(J6, J3:J6,O3:O6,"NA",0,1)				
13	Input the lookup values in group						
14		Input	Result				
15							
16		Roll Number	Total				
17		J3:J6	85	=XLOOKUP(J3:J6,J3:J6,O3:O6,"NA",0,1)			
18			81				
19			89				
20			93				

Lookup for single value say J3
 =XLOOKUP(J3, J3:J6, O3:O6, "NA", 0, 1)

Lookup for a group of values J3:J6
 =XLOOKUP(J3:J6, J3:J6, O3:O6, "NA", 0, 1)

Referring Cells Using XLOOKUP - 05



	A	B	C	D	E	F	G
1	Horizontal lookup - one value lookup at a time	Lookup Table					
2		Date	1/1/2005	5/1/2005	8/1/2005		
3		Price	98	105	112		
4							
5							
6							
7							
8							
9							
10							
11							
12							

Lookup for single value say C10
 =XLOOKUP(C10, C2:E2, C3:E3, "NA", -1)

Input	Result
Date	Price
1/4/2005	98
5/10/2005	105
9/10/2005	112
5/1/2005	105

=XLOOKUP(C8,C2:E2,C3:E3,"NA",-1)
 =XLOOKUP(C9,C2:E2,C3:E3,"NA",-1) vs
 =XLOOKUP(C10,C2:E2,C3:E3,"NA",-1)
 =XLOOKUP(C11,C2:E2,C3:E3,"NA",-1)



Referring Cells Using XLOOKUP - 06

	A	B	C	D	E	F	G
1	Horizontal lookup - one value lookup at a time	Lookup Table					
2		Date	1/1/2005	5/1/2005	8/1/2005		
3		Price	98	105	112		
4							
5							
6			Input	Result			
7			Date	Price			
8			1/4/2005	98	=XLOOKUP(C8,C2:E2,C3:E3,"NA",-1)		
9			5/10/2005	105	=XLOOKUP(C9,C2:E2,C3:E3,"NA",-1) vs		
10			9/10/2005	112	=XLOOKUP(C10,C2:E2,C3:E3,"NA",-1)		
11			5/1/2005	105	=XLOOKUP(C11,C2:E2,C3:E3,"NA",-1)		
12							
13							
14							
15			Group of inputs C8:C11	98	=XLOOKUP(C8:C11,C2:E2,C3:E3,, -1)		
16				105			
17				112			
18				105			

Lookup for single value say C10
=XLOOKUP(C8:C11, C2:E2, C3:E3, "NA", -1)

Referring Cells Using XLOOKUP - 07



	D	E	F	G	H	I	J
1							
2							
3							
4							
5	Income	Tax rate					
6	0	0.15					
7	10000	0.3					
8	30000	0.34					
9	100000	0.4					
10							
11	Input	Result					
12	Income	Rate					
13	-1000	NA	=XLOOKUP(D13, D6:D10, E6:E10, "NA", 0)		Match mode	0	Exact match
14	-1000	NA	=XLOOKUP(D14, D6:D9, E6:E9, "NA", -1)		Match mode	-1	Exact match or next smaller
15	-1000	0.15	=XLOOKUP(D15, D6:D9, E6:E9, "NA", 1)		Match mode	1	Exact match or next larger
16							



LOOKUP Function – Analytics – Assign Grades



VLOOKUP – Analytics – Assigning Grade



D1							Roll Number
	A	B	C	D	E	F	G
1	Marks Range	Letter Grade		Roll Number	Marks	Grade	
2	0	F		1234	87	AB	=VLOOKUP(E2,Ig, 2,1)
3	30	DD		1235	91	AA	=VLOOKUP(E3,Ig, 2,1)
4	40	CD		1236	77	BB	=VLOOKUP(E4,Ig, 2,1)
5	50	CC		1237	47	CD	=VLOOKUP(E5,Ig, 2,1)
6	60	BC		1238	58	CC	=VLOOKUP(E6,Ig, 2,1)
7	70	BB		1239	63	BC	=VLOOKUP(E7,Ig, 2,1)
8	80	AB		1240	24	F	=VLOOKUP(E8,Ig, 2,1)
9	90	AA					
10							





INDEX Function





Referring Cells Using INDEX - 01

- Group of cells are referred using colon notation
- A1:C3 refers to cells A1, A2, A3, B1, B2, B3, C1, C2 and C3
- Index function allows to us to think group of cells A1:C3 as if they are a two-dimensional array (or a matrix).
- Access the elements using row number and column number notation

	A (1)	B (2)	C (3)
1	A1	B1	C1
2	A2	B2	C2
3	A3	B3	C3

B2 is accessed as: INDEX(A1:C3, 2, 2)

C2 is accessed as: INDEX(A1:C3, 2, 3)



INDEX Function – Analytics - Compute Trip Distance



Referring Cells Using INDEX - 02



- A basketball team is playing matches in eight cities as shown.
- They begin from Seattle, plays at Boston, Chicago, Dallas, Denver, LA, Miami, Phoenix and come back to Seattle
- Distance between pair of cities is given. Compute total trip in KM covered by Seattle team for the above trip.
- That is SUM of distance between
 - Seattle to Phoenix
 - Phoenix to LA
 - LA to Denver
 - Denver to Dallas
 - Dallas to Chicago
 - Chicago to Seattle

Referring Cells Using INDEX - 03



B29										
	A	B	C	D	E	F	G	H	I	J
3										
4		Boston-De	1991	=INDEX(C10:J17, 1,4)		T Dist to Seattle				
5		Seattle- M	3389	=INDEX(C10:J17,8,6)						
6										
7										
8										
9			Boston	Chicago	Dallas	Denver	LA	Miami	Phoenix	Seattle
10	1	Boston	0	983	1815	1991	3036	1539	2664	2612
11	2	Chicago	983	0	1205	1050	2112	1390	1729	2052
12	3	Dallas	1815	1205	0	801	1425	1332	1027	2404
13	4	Denver	1991	1050	801	0	1174	2100	836	1373
14	5	LA	3036	2112	1425	1174	0	2757	398	1909
15	6	Miami	1539	1390	1332	2100	2757	0	2359	3389
16	7	Phoenix	2664	1729	1027	836	398	2359	0	1482
17	8	Seattle	2612	2052	2404	1373	1909	3389	1482	0
18										
19			Road Trip!!							
20			City	Distance						
21		Seattle	8							
22		Phoenix	7	1482	=INDEX(distance, C21, C22)					
23		LA	5	398	=INDEX(distance, C22, C23)					
24		Denver	4	1174	=INDEX(distance, C23, C24)					
25		Dallas	3	801	=INDEX(distance, C24, C25)					
26		Chicago	2	1205	=INDEX(distance, C25, C26)					
27		Seattle	8	2052	=INDEX(distance, C26, C27)					
28			Total	7112	=SUM(D22:D27)					





MATCH Function



MATCH Function - 01



- Column A has list of roll numbers.
- Given a specific roll number 1234, find in which row this value is appearing in column A?
- In this example, this roll number is appearing in second row.
- This example shows an exact match of **MATCH**

	Roll Number
1	1233
2	1234
3	1235
4	1236

MATCH Function - 02



- Approximate match example
- Column A has list of roll numbers.
- Given a specific roll number 1238, find the largest roll number that is smaller than 1238.
- In this example, this roll number is appearing in third row.
- This example shows an exact match of MATCH

	Roll Number
1	1233
2	1235
3	1237
4	1239



MATCH Function – Analytics – Sales of Product During Specified Month



MATCH Function - 03



- Given monthly sales of several products.
- Compute the sales of specific product in the given month

56

MATCH Function - 05



- Given the list of baseball players and their salaries.
- Find the player with highest salary.
- Use MAX function to find highest salary
- Use MATCH function to determine the row of the highest salary
- Use VLOOKUP to find player name.

MATCH Function - 06



C8					
	A	B	C	D	E
1					
2					
3					
4			highest	Formula	5th highest
5	player position	345	=MATCH(MAX(salaries), salaries, 0)	232	=MATCH(LARGE(salaries,5),salaries,0)
6	amount	22000000	=MAX(salaries)	12600000	=LARGE(salaries,5)
7	name	Alex Rodriguez	=VLOOKUP(C5, A13:B413,2,FALSE)	dl-Derek Jeter	=VLOOKUP(E5,A13:B413, 2, 0)
8					
9					
10					
11					
12		name	salary		
13	1	dl-Mo Vaughn	13166667		
14	2	Tim Salmon	5683013		
15	3	Garret Anderson	4500000		
16	4	Darin Erstad	3450000		
17	5	Troy Percival	3400000		
18	6	Ismael Valdes	2500000		
19	7	Pat Rapp	2000000		
20	8	Glenallen Hill	1500000		
21	9	Troy Glaus	1250000		
22	10	Shigetoshi Hasegawa	1150000		
23	11	Scott Spiezio	1125000		
24	12	Orlando Palmeira	900000		



Thank You!

